

Creating AERO.inp for the FS

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What you need

- Windows OS (preferred, but can be done without it)
 - RASAero II installed
 - Python3 installed
- brute_force_aero.py
- MatLab
 - InputFileCreator.m
 - InputFileAppend.m

Understanding the AERO input file for the FS

The input file looks like the following:

alpha	mach	xCP	clCoast	cdCoast	clBoost	cdBoost			
0.000000		0.010000		116.053000		0.000000	0.319000	0.000000	0.315000
0.000000		0.020000		116.053000		0.000000	0.314000	0.000000	0.311000
0.000000		0.030000		116.053000		0.000000	0.303000	0.000000	0.299000
0.000000		0.040000		116.053000		0.000000	0.293000	0.000000	0.289000
0.000000		0.050000		116.053000		0.000000	0.285000	0.000000	0.281000
0.000000		0.060000		116.053000		0.000000	0.278000	0.000000	0.274000
0.000000		0.070000		116.053000		0.000000	0.272000	0.000000	0.268000
0.000000		0.080000		116.053000		0.000000	0.267000	0.000000	0.263000
0.000000		0.090000		116.053000		0.000000	0.263000	0.000000	0.259000
0.000000		0.100000		116.053000		0.000000	0.259000	0.000000	0.255000

The file varies from alpha (angle of attack) 0 to 30 (increments of 1) and from Mach 0.01 to 1.5 (increments of 0.01). At each alpha and Mach combination, is an accompanying value for:

- xCP
 - position of center of pressure from nosecone (inches)
- clCoast
 - Coefficient of lift with engine off
- cdCoast
 - Coefficient of drag with engine off
- clBoost
 - Coefficient of lift with engine on
- cdBoost
 - Coefficient of drag with engine on

Creating the AERO input file

The first step to creating the AERO input file is creating 31 .txt files containing the CL and CD values for a single alpha value at each Mach number. We will call these files alpha files. An alpha file for alpha=16 would look like the following:

```

-----> BODY TRANSITIONING TO TURBULENT FLOW <-----

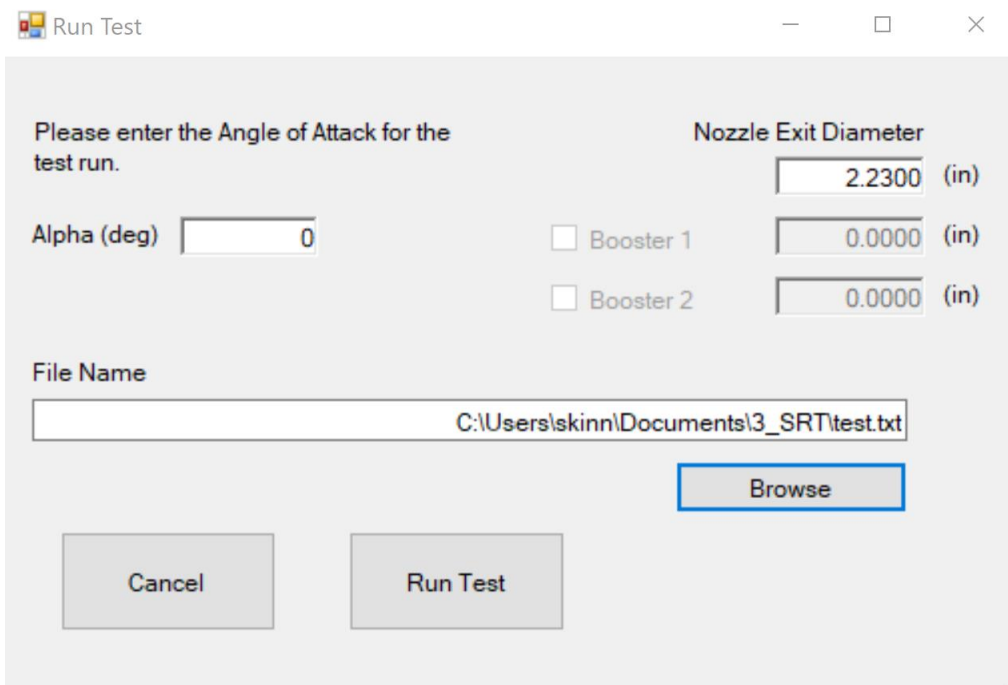
0.01  0.00  0.319  0.315  0.191  0.011  0.018  0.057  0.042  0.000  0.000  962313
0.01  9.33 116.053
0.01 16.00  2.605  0.000
0.01 16.00  2.605 116.053
0.01 16.00  2.409  1.050  2.605  0.345
0.01 16.00  2.410  1.046  2.605  0.341

0.02  0.00  0.314  0.311  0.215  0.012  0.017  0.041  0.030  0.000  0.000  1924626
0.02  9.33 116.053
0.02 16.00  2.605  0.000
0.02 16.00  2.605 116.053
0.02 16.00  2.410  1.045  2.605  0.340
0.02 16.00  2.412  1.041  2.605  0.336

```

To create the alpha files, you will need to use the brute_force_aero.py script. Instructions on using the script are as follows:

1. Open RASAero II
 - a. Load your .CDX1 file
2. Go to tools > run test. If the following window pops up, you are good to continue. If the window does not pop up, consult the RASAero user's guide to troubleshoot possible issues. Do not worry about changing the file path to save test files to, this will be done later.



The image shows a 'Run Test' dialog box with the following fields and controls:

- Please enter the Angle of Attack for the test run.**
 - Alpha (deg):
- Nozzle Exit Diameter**
 - (in)
- Booster 1**
 - ☐ (in)
- Booster 2**
 - ☐ (in)
- File Name**
 -
 -
-

3. Open brute_force_aero.py
4. Change aero_path to a folder in your computer that can contain the 31 alpha files once created.
5. Open a command prompt
 - a. Note: It is highly recommended you run the .py file in your command prompt to avoid IDE hotkeys interfering with the hotkeys used by the script

6. Change directory to the location of the .py file. **Do not run yet!**
7. Go back to the RASAero ii window, then press “alt + tab”. If your window switches back to the command prompt, you are ready to run the .py file. If you do not switch back to the command prompt, troubleshoot your windows tabbing hot keys.
8. Run the .py file

Following the above steps will create 31 alpha files that are ready to be merged into a single aero input file. Use the following steps to complete that merging:

1. Open InputFileCreator.m
 - a. Make sure InputFileCreator.m and InputFileAppend.m is in the same folder as all 31 alpha files
2. Change filename in the InputFileCreator.m
3. Run InputFileCreator.m

You will now have one compiled AERO input file. This input file is ready for use in the FS.